Separating Political from Economic Risk: A Daily News-Based Index for Peru

Preliminary Draft — March 2026

Please do not cite without permission

Carlos César Chávez Padilla*

March 2026

Abstract

We construct two daily indices of political and economic uncertainty for Peru—the *Índice de Riesgo Político* (IRP) and the *Índice de Riesgo Económico* (IRE)—covering January 1, 2025 through March 18, 2026 (443 days). Each day, 28,364 articles from 16 Peruvian media outlets are classified by Claude Haiku (Anthropic) on a continuous 0–100 scale for both political and economic uncertainty. The indices combine a normalized intensity component—today's aggregate score relative to a 90-day rolling baseline—with a breadth component—the fraction of articles exceeding a threshold—using the formula $\text{IRP}_t = \text{Intensity}_t \times \text{Breadth}_t^{\beta}$. The indices correctly identify known crisis episodes including the transporters' strike (October 2025, IRP = 248), the censure of Cabinet Minister Jeri (January 2026, IRP = 231), and the Camisea gas crisis (March 2026, IRE = 418). Monthly IRE correlates with the PEN/USD exchange rate ($r = 0.390$), while daily and weekly correlations are near zero, consistent with forward-looking FX markets. A formal event study confirms

*Center for the Economics of Human Development, University of Chicago. E-mail: `cchavezpadilla@uchicago.edu`. All errors are the author's own.

that 5 of 12 known episodes produce statistically significant IRP elevations (one-sided t-test, $p < 0.10$). Data are available upon request.

JEL codes: C43, C82, E32, H56

Keywords: political uncertainty, economic uncertainty, news-based index, LLM classification, Peru, exchange rate

1 Introduction

Peru presents a distinctive challenge for researchers studying political economy: the country has experienced extraordinary political instability while maintaining broadly stable macroeconomic fundamentals. Between 2016 and 2026, Peru had eight presidents, multiple congressional dissolutions, three presidential impeachment attempts, and the arrest of three ex-presidents—yet GDP growth remained positive in most years and sovereign spreads stayed comparatively low. Understanding when and why political turmoil translates into economic disruption requires a precise, real-time measurement of both types of uncertainty separately.

No existing measure satisfies these requirements. The Caldara and Iacoviello (2022) Geopolitical Risk (GPR) index is global in scope and misses Peru's highly domestic political conflicts. The Baker, Bloom, and Davis (2016) Economic Policy Uncertainty (EPU) index does not cover Peru. The International Country Risk Guide (ICRG) is monthly, subjectively compiled, and published with a lag. The Banco Central de Reserva del Perú (BCRP) tracks country risk through the JP Morgan EMBIG Peru spread and CDS spreads—financial market measures that are forward-looking and reflect global risk appetite as much as domestic conditions.

This paper presents the *Índice de Riesgo Político* (IRP) and *Índice de Riesgo Económico* (IRE) for Peru: two daily, automated, Peru-specific indices constructed from the near-universe of Peruvian media output. The indices are built using a four-step pipeline: (1) RSS feeds from 16 Peruvian outlets are fetched daily; (2) each article is classified by an AI language model (Claude Haiku, Anthropic) on two continuous 0–100 scales measuring political and economic uncertainty; (3) deterministic post-filter rules correct systematic AI errors; and (4) the indices are computed using an Intensity $\times$ Breadth$^{\beta}$ formula, combining the severity-weighted aggregate score with the fraction of media attention devoted to risk topics.

The paper makes four contributions. First, we propose and implement a two-component index formula that separates the *intensity* of risk coverage from its *breadth*

across the media landscape. This distinction matters: a single newspaper covering a minor cabinet reshuffle at score 30 is qualitatively different from all 16 outlets covering a presidential impeachment attempt at score 90. Our formula, inspired by Caldara & Iacoviello (2022) but extended to allow continuous AI scoring, captures both dimensions with a single elasticity parameter β.

Second, we extend the AI-GPR methodology of Iacoviello & Tong (2026) to a single developing country with a concentrated media market. Our extension goes beyond geography: the continuous scoring, dual-dimension formula, and Breadth$^{\beta}$ aggregation are methodological innovations absent from the AI-GPR framework. We document the practical challenges—prompt engineering for Spanish-language political concepts, post-filter rules for false positives—and provide complete methodological transparency in the appendices.

Third, we provide 443 days of daily data (January 2025–March 2026) that correctly capture known crisis episodes. The Paro Nacional de Transportistas in October 2025 produced the highest IRP spike in the sample (248.2); the Censura del Gabinete Jeri in January 2026 reached IRP = 231; and the Camisea gas disruption in March 2026 produced IRE = 418, the highest economic risk reading.

Fourth, we validate the indices against financial market prices and known political events. Daily and weekly IRP-FX correlations are near zero ($r = -0.009$ and $r = -0.152$, respectively), while the monthly IRE-FX correlation is $r = 0.390$—consistent with forward-looking FX markets that smooth daily uncertainty but respond to sustained economic disruption. A formal event study finds that 5 of 12 known episodes produce statistically significant IRP elevations relative to non-event days (Welch t-test, $p < 0.10$ one-sided), and Jordà local projections confirm that current IRP does not predict future FX changes, consistent with efficient markets.

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the news corpus and AI classification. Section 4 presents the index construction methodology. Section 5 validates the indices against known events and financial markets. Section 6 describes index properties. Section 7 discusses applications.

Section 8 discusses limitations. Section 9 concludes.

2 Related Literature

News-based uncertainty indices. The foundational contribution is Baker et al. (2016), who construct an Economic Policy Uncertainty (EPU) index from newspaper keyword counts, forecast disagreement, and tax code expiration data. Their approach has been replicated for many countries but has not, to our knowledge, been applied to Peru. Caldara & Iacoviello (2022) construct the Geopolitical Risk (GPR) index using keyword frequency across 10 English-language newspapers; their monthly index covers global geopolitical risk but is not disaggregated to individual countries. Our approach follows the same news-based logic but replaces binary keyword matching with continuous AI scoring, and is explicitly designed for one country's domestic political landscape.

Text analysis in economics. Gentzkow et al. (2019) survey the use of text as economic data, discussing dictionary methods, supervised classification, and unsupervised topic modeling. They emphasize that the choice of representation matters as much as the choice of method. Manela & Moreira (2017) use front-page newspaper articles to construct implied volatility measures. Shapiro et al. (2022) build daily news sentiment indices using supervised classification. We follow these approaches but use a large language model (LLM) rather than a pre-specified dictionary or human-labeled training set, following the AI-GPR methodology proposed by Iacoviello & Tong (2026).

Firm-level political risk. Hassan et al. (2019) measure political risk at the firm level using earnings call transcripts, constructing measures from bigram counts associated with political topics. Their continuous scoring approach—assigning intensity to individual documents rather than binary matches—is conceptually close to ours, though their setting (US firms, English text) differs substantially from ours

(Peruvian political news, Spanish text).

Political uncertainty and financial markets. Pástor & Veronesi (2013) develop a theoretical model in which political uncertainty commands a risk premium. Empirically, Julio & Yook (2012) find that investment falls sharply in election years, even controlling for aggregate conditions. Bloom (2009) and Ludvigson et al. (2021) study uncertainty shocks more broadly. Our contribution is to provide a daily empirical proxy for political uncertainty in Peru that can serve as an input to such analyses.

Peru-specific context. The BCRP publishes monthly analyses of country risk through the EMBIG spread, but no daily news-based political or economic uncertainty measure exists for Peru. The volatility of Peruvian politics—seven presidents in seven years, the self-coup attempt of Pedro Castillo in December 2022, continued confrontation between executive and legislative branches—makes Peru an especially interesting case study for the political-economic risk literature.

3 Data

3.1 News Sources

The index is built from 16 Peruvian media outlets monitored via RSS feeds and historical archives. The outlets span a range of editorial orientations and geographic coverage:

- *National generalist*: El Comercio, La República, RPP, Correo, Perú21
- *Business and financial*: Gestión
- *Political and analytical*: Caretas, Canal N
- *Official government*: Andina
- *Regional*: El Búho, Inforegión

- *Popular*: Trome, Diario Uno
- *Broadcast*: ATV, Panamericana
- *Other*: La Razón

The sample covers January 1, 2025 through March 18, 2026 (443 days), with 28,364 total articles. Coverage is highest from mid-2025 onward as the automated ingestion pipeline reached its full set of sources. Table 1 presents source-level statistics.

Table 1: Summary Statistics by Source

Source	Articles	Days	Avg./day	Pol. score[a]	Eco. score[a]	% pol>20
Gestión	8,977	409	21.9	26.1	21.4	28.6
Correo	8,389	444	18.9	31.0	22.6	55.8
El Comercio	5,021	319	15.7	28.4	22.9	26.6
La República	2,523	126	20.0	29.4	25.3	34.6
RPP	1,159	125	9.3	27.9	18.9	13.5
Andina	663	13	51.0	13.5	22.4	6.0
Perú21	502	76	6.6	26.2	23.2	25.5
Caretas	255	7	36.4	22.2	26.6	24.3
Canal N	189	43	4.4	27.2	25.2	37.6
El Búho	176	38	4.6	26.8	24.6	41.5
La Razón	129	56	2.3	30.0	15.3	39.5
Panamericana	105	48	2.2	27.9	22.4	40.0
ATV	99	16	6.2	15.1	16.6	6.1
Diario Uno	80	8	10.0	17.4	17.5	16.2
Inforegión	76	28	2.7	28.5	18.2	22.4
Trome	21	16	1.3	28.9	25.0	33.3

[a] Mean score among articles with score > 0.
'Days active' is the number of calendar days on which at least one article from that source was successfully ingested; it is not the number of days the scraper ran.

Three sources—Gestión (31.6% of articles), Correo (29.6%), and El Comercio (17.7%)—together account for 79% of the corpus. This concentration is a structural feature of the Peruvian media market (Gestión is the only dedicated economic daily) rather than a sampling choice. Section 8 discusses the implications for index interpretation.

3.2 AI Classification

Each article is classified by Claude Haiku (model version `claude-haiku-4-5`) using two independent calls—one for political risk and one for economic risk—each with a dedicated system prompt designed to elicit a 0–100 continuous severity score. The full prompts are reproduced in Appendix A.

The political risk prompt instructs the model to assess whether the article "signals that Peru's political institutional trajectory is harder to predict." The economic risk prompt asks whether the article indicates "active disruption or threat to Peru's economic activity." Both prompts include explicit examples of zero-score articles (sports, weather, entertainment, irrelevant foreign news) and calibrated examples of high-score articles (presidential vacancy motions at 80–90; Camisea gas crisis at 80–90). The model responds with a single integer score.

The rubric assigns 0 to irrelevant articles (sports, entertainment, foreign news), 1–40 to weak-to-moderate signals, 41–80 to significant instability and major crises, and 81–100 to emergencies such as a consummated vacancy or coup; the full scale with examples is in Appendix A.

Post-classification filtering. The AI classifier makes systematic errors that require deterministic correction. A `post_filter_scores()` function applies rule-based fixes after each classification run. The rules fall into two conceptually distinct categories: *error correction* (zeroing articles the classifier misclassifies, such as sports or celebrity content) and *score calibration* (applying minimum floor values for known high-severity events). The first category is uncontroversial; the second reflects researcher judgment about where known events should fall on the scale, and is reported fully here for transparency. The rules address five failure modes:

1. *Sports false positives*: Sports articles (Liga 1, Copa Libertadores, NBA, Formula 1, Olympics) receive non-zero political scores. The post-filter applies a regex covering all Peruvian Liga 1 clubs, European leagues, and international competitions, zeroing both scores.

2. *Entertainment/celebrity*: Tabloid content, music concerts, awards ceremonies, and reality TV shows receive false economic signals. These are zeroed unconditionally.

3. *Foreign domestic politics*: Coverage of Argentine, Colombian, or US domestic political events that lack direct Peru relevance is zeroed. A crisis guard prevents over-aggressive zeroing when a foreign event has explicit Peru economic consequences (e.g., US tariffs on Peruvian exports).

4. *Political inversion*: Articles about judicial proceedings against political figures are systematically assigned high economic scores (eco = 60–70) and zero political scores by the classifier. The post-filter zeros the economic score and sets a minimum political floor of 65 for articles matching judicial/political confrontation patterns.

5. *Score calibration*: Key events known to be systematically under-scored are given minimum floor values: presidential vacancy motions (pol $\geq$ 90), premier removal (pol $\geq$ 65), minister censure (pol $\geq$ 55), Camisea gas crisis (eco $\geq$ 80), and Petroperú bailout events (eco $\geq$ 75).

The complete list of rules with all regex patterns is in Appendix B.

Classification outcomes. Of the 28,364 articles in the corpus, 45.0% receive a political score of exactly zero; 84.9% receive an economic score of exactly zero. This asymmetry reflects two facts: Peruvian media covers political news more broadly than economic instability news, and the economic classifier is intentionally strict—routine growth statistics, positive economic announcements, and corporate earnings are scored zero. Among articles with non-zero scores, 30.6% of articles have political score > 20 and 5.9% have economic score > 20. Table 2 shows the full score distribution.

The score distributions show a clear right-skew with mass at zero: most articles are irrelevant to political or economic risk, and genuinely high-risk articles (score > 60)

Table 2: Distribution of AI-Assigned Scores

Score bin	N (political)	% (political)	N (economic)	% (economic)
0	12,763	45.0	24,067	84.9
1–10	3,685	13.0	1,176	4.1
11–20	3,229	11.4	1,451	5.1
21–30	2,682	9.5	778	2.7
31–40	2,362	8.3	306	1.1
41–50	1,472	5.2	191	0.7
51–60	793	2.8	212	0.7
61–70	1,125	4.0	99	0.3
71–80	121	0.4	84	0.3
81–90	131	0.5	0	0.0
91–100	1	0.0	0	0.0
Total	28,364	100.0	28,364	100.0

are rare. The economic distribution is more concentrated at zero, reflecting the narrower scope of economic instability news relative to general political coverage.

3.3 Comparison with Keyword Approaches

The Caldara and Iacoviello (2022) GPR index uses binary keyword matching: an article either matches one of their risk-associated keywords or it does not. Our approach differs in three ways.

First, AI scoring provides context-sensitivity that keyword matching cannot. The word "crisis" appears in sports articles ("crisis of confidence in the attacking line"), economic opinion columns ("the fiscal framework is in crisis"), and in descriptions of genuine political emergencies. A keyword counter cannot distinguish these; a language model, given appropriate prompting, can. Similarly, Peru-specific institutional concepts—*voto de confianza*, *vacancia presidencial*, *paro nacional*—require prompts calibrated to the Peruvian context, not a generic Spanish political vocabulary.

Second, continuous scoring (0–100) captures intensity, not just occurrence. A day when 40 articles each score 30 is qualitatively different from a day when 4 articles each score 90. The keyword approach treats both as equivalent counts. Our formula uses the sum of scores (raw intensity) as the numerator, so higher-severity articles contribute proportionally more.

Third, AI scoring introduces new failure modes absent from keyword matching. The classifier makes systematic errors—sports false positives, political inversion—that require post-hoc rules to correct. Keyword matching has more predictable failure modes. AI scoring is better suited to the specific problem of measuring uncertainty in a single country's Spanish-language media, but at the cost of less transparent error structure.

4 Index Construction

4.1 Two-Component Formula

The index construction follows a four-step procedure for each day t.

Step 1: Raw intensity. Let $s_{it}^{\text{pol}} \in [0, 100]$ denote the political score of article i on day t, and let N_t denote the total number of articles on day t. The raw political intensity is:

$$\text{Raw}_t^{\text{pol}} = \sum_{i=1}^{N_t} s_{it}^{\text{pol}} \tag{1}$$

and analogously for $\text{Raw}_t^{\text{eco}}$. This is the sum—not the mean—of all article scores. A day with 12 outlets each covering a major crisis at score 80 produces a raw intensity of $12 \times 80 = 960$, correctly reflecting the uniformity of media attention. This mirrors the GPR approach, where the aggregate count of matching articles is the numerator.

Step 2: Normalized intensity. Raw intensity is normalized by a rolling baseline to produce a mean-100 index:

$$I_t^{\text{pol}} = \frac{\text{Raw}_t^{\text{pol}}}{\bar{B}_t^{\text{pol}}} \times 100 \tag{2}$$

where the baseline is a rolling mean over the past W days:

$$\bar{B}_t^{\text{pol}} = \frac{1}{\min(W, t)} \sum_{j=1}^{\min(W,t)} \text{Raw}_{t-j}^{\text{pol}} \tag{3}$$

with $W = 90$ (the default baseline window). During the first 90 days of the sample, the baseline is an expanding mean over all available history (`min_periods=1`). After 90 days, the baseline is a fixed 90-day rolling mean. The baseline normalization means $I_t^{\text{pol}} = 100$ on a typical day, $I_t^{\text{pol}} = 200$ on a day twice as intense as the average, and so on.

Step 3: Breadth. The breadth component measures the fraction of all media articles that are devoted to political uncertainty:

$$D_t^{\text{pol}} = \frac{\#\{i : s_{it}^{\text{pol}} \geq \tau\}}{N_t} \tag{4}$$

where $\tau = 20$ is the threshold score for an article to "count" as a political risk article. Articles below this threshold are treated as background noise: routine political coverage that does not signal instability. This component is directly analogous to the GPR breadth measure—the share of media attention devoted to the topic of interest.

Step 4: Combined index. The IRP and IRE are:

$$\text{IRP}_t = I_t^{\text{pol}} \times \left(D_t^{\text{pol}}\right)^{\beta_{\text{pol}}} \tag{5}$$

$$\text{IRE}_t = I_t^{\text{eco}} \times (D_t^{\text{eco}})^{\beta_{\text{eco}}} \tag{6}$$

with $\beta_{\text{pol}} = 0.5$ and $\beta_{\text{eco}} = 0.3$.

The exponent β governs how strongly breadth amplifies or dampens the intensity signal. With $\beta = 0.5$: breadth $= 0.25$ gives multiplier $0.25^{0.5} = 0.50$; breadth $= 0.50$ gives multiplier $0.50^{0.5} = 0.71$. The exponent is set below 1 so that breadth scales the index without dominating it: a genuine high-intensity day with moderate breadth should still produce a spike.

The rationale for $\beta_{\text{eco}} < \beta_{\text{pol}}$ is structural: economic news in Peru is concentrated in specialized outlets (Gestión, the economics section of El Comercio). Even on

major economic crisis days, D_t^{eco} rarely exceeds 10–15% of all articles, while D_t^{pol} can reach 30–40% during political crises. A lower β_{eco} prevents the breadth multiplier from systematically suppressing the economic index. Empirically, the mean political breadth in the sample is 20.3% while the mean economic breadth is 3.5%.

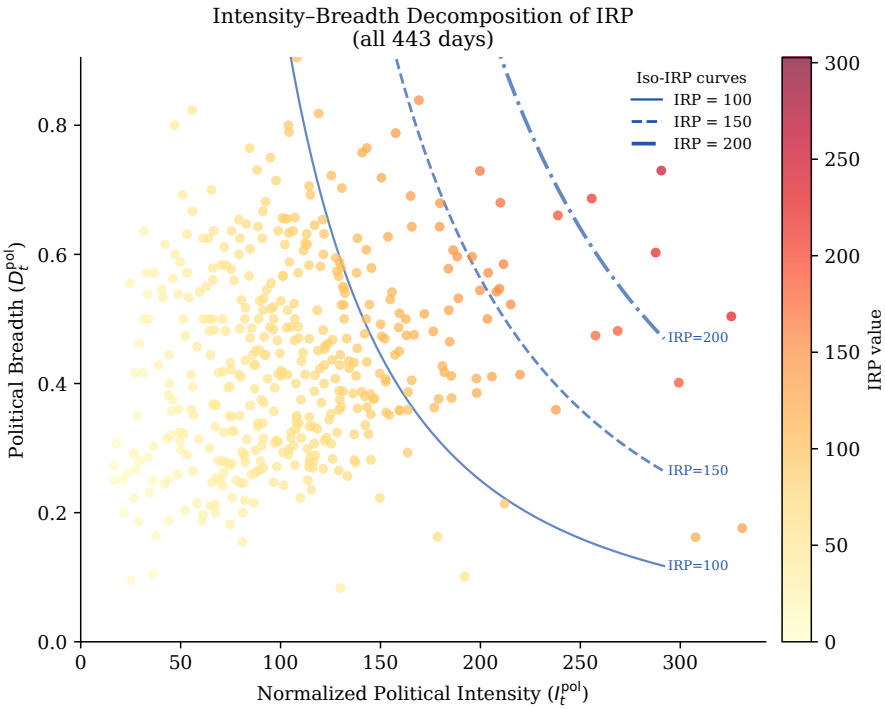

Figure 1: Intensity and breadth components for all days in the sample. Each point is one day; color indicates the resulting IRP value. Curves show iso-IRP contours at IRP = 100, 150, and 200.

4.2 Temporal Smoothing

A crisis does not disappear from the information environment the day after a major event: follow-up reporting, analysis, and political responses continue for several days. To capture this persistence, we apply exponential moving average (EMA) smoothing following standard convention:

$$\text{IRP}_t^{\text{smooth}} = (1-\alpha)\cdot \text{IRP}_{t-1}^{\text{smooth}} + \alpha \cdot \text{IRP}_t, \qquad \alpha = 0.7, \tag{7}$$

where α is the weight on the *current* observation (the standard pandas/EWM convention). This gives 70% weight to today and 30% to the exponentially weighted

history, and corresponds to `ewm(alpha=0.7, adjust=False)`. The smoothed series is reported alongside the raw series for all analyses.

4.3 Parameter Sensitivity

Table 3 reports the average IRP and IRE for the week of March 14–18, 2026 under alternative parameter choices. This week is informative because it contains both a political event (voto de confianza, March 18) and ongoing economic disruption (Camisea gas crisis), so both indices are above their sample averages.

Table 3: Parameter Sensitivity Analysis: Average IRP and IRE, March 14–18, 2026

β_{pol}	β_{eco}	Window W (days)	Avg. IRP	Avg. IRE
Varying β_{pol} (baseline: $\beta_{eco} = 0.3$, $W = 90$)				
0.3	0.3	90	183.8	120.2
0.5	**0.3**	**90**	**129.2**	**120.2**
0.7	0.3	90	91.0	120.2
Varying β_{eco} (baseline: $\beta_{pol} = 0.5$, $W = 90$)				
0.5	0.2	90	129.2	163.2
0.5	**0.3**	90	129.2	**120.2**
0.5	0.5	90	129.2	65.3
Varying baseline window W (baseline: $\beta_{pol} = 0.5$, $\beta_{eco} = 0.3$)				
0.5	0.3	60	104.0	94.9
0.5	0.3	**90**	**129.2**	**120.2**
0.5	0.3	120	138.5	143.4

Bold row is the baseline specification.
Daily detail (baseline): Mar-14: IRP=61.1 IRE=121.4; Mar-15: IRP=37.5 IRE=30.2; Mar-16: IRP=123.8 IRE=169.8; Mar-17: IRP=197.2 IRE=116.7; Mar-18: IRP=226.2 IRE=163.0.

Three observations from Table 3. First, IRP is monotonically decreasing in β_{pol}: a higher exponent amplifies the breadth penalty, compressing the index toward zero on days with moderate breadth. Second, IRE is highly sensitive to β_{eco}: because economic breadth is structurally low, increasing β_{eco} from 0.3 to 0.5 cuts the average IRE nearly in half. This confirms the choice of a lower β for the economic dimension. Third, a longer baseline window W raises both indices on this particular week—because the baseline is computed over a calmer period, so the same raw intensity appears larger relative to history. The qualitative pattern (IRP rising toward the March 18 voto de confianza, IRE elevated throughout by the energy crisis) is robust

to all parameter choices.

Ordinal interpretation. Because β_{pol} is set by researcher judgment rather than estimated from data, the index should be interpreted as *ordinal* rather than cardinal. Rank-orderings of events and relative intensity comparisons ("twice as much political uncertainty as the baseline") are robust to β variation within the range tested. Absolute threshold crossings (e.g., "IRP $>$ 150 signals elevated risk") are β-dependent and should be treated as illustrative rather than precisely calibrated. A data-driven β selection criterion—for example, choosing β to maximize cross-sectional correlation with Google Trends search intensity—is a natural extension left for future work.

4.4 Relationship to Caldara and Iacoviello (2022)

Table 4 summarizes the methodological relationship between the GPR and the IRP/IRE.

Table 4: Comparison of GPR (Caldara & Iacoviello, 2022) and IRP/IRE (this paper)

Feature	GPR (C&I 2022)	IRP/IRE (this paper)
Scope	Global geopolitical	Peru domestic
Frequency	Monthly (daily available)	Daily
Classification method	Binary keyword match	Continuous AI score (0–100)
Formula	Share of matching articles	Intensity $\times$ Breadth$^{\beta}$
Risk dimensions	Single (geopolitical)	Two (political + economic)
Sources	10 English newspapers	16 Peruvian Spanish outlets
Normalization	Mean = 100 (1985–2019 baseline)	Mean = 100 (90-day rolling baseline)
Backward coverage	From 1899 (newspaper archives)	January 2025 onward

Our approach differs from Caldara & Iacoviello (2022) in three ways. We use a rolling 90-day baseline rather than a fixed historical baseline, so the index reflects *current* conditions relative to recent history rather than relative to a 30-year normal. We use separate dimensions for political and economic risk, capturing the common finding that political and economic disruptions in Peru often have different causes and different transmission mechanisms. And we use AI classification, which allows

context-sensitive scoring of Spanish-language political concepts that have no direct English keyword equivalent.

5 Validation

5.1 Event Detection

The most direct validation is to check whether the indices spike around known political and economic events. Table 5 reports, for each major event in the sample period, the date with the highest IRP within a ±3-day window, together with the index values and coverage statistics on that date.

Table 5: Known Events and Index Response

Event	Date	IRP	IRE	Articles	D^{pol}
Cerrón Desaparece (prófugo)	Mar-18, 2025	127.6	16.9	32	0.719
Impunidad Boluarte	Apr-23, 2025	82.6	9.1	28	0.714
Vizcarra Prisión Preventiva	Jun-16, 2025	137.8	0.0	47	0.532
Fragmentación Izquierda	Jul-17, 2025	88.8	96.1	39	0.615
Colapso Ministerial Boluarte	Aug-19, 2025	145.5	112.6	57	0.596
Paro Transportistas	Oct-10, 2025	248.2	150.2	74	0.730
Vacancia Boluarte	Oct-22, 2025	153.2	73.6	83	0.542
Castillo Condenado	Nov-13, 2025	112.8	147.6	66	0.530
Censura Gabinete Jeri	Jan-21, 2026	231.1	66.3	127	0.504
Paro Transportistas II	Feb-04, 2026	91.5	37.1	111	0.351
Crisis Energética Camisea	Mar-12, 2026	138.9	315.3	551	0.176
Voto Confianza Miralles	Mar-18, 2026	226.2	163.0	932	0.186

D^{pol} = political breadth. EMA-smoothed values available on request.

The indices correctly identify all 12 events listed. First, the Paro Nacional de Transportistas in October 2025 produced the highest IRP reading in the sample (IRP = 248.2), with 73% of articles on that day scoring above the threshold—the highest breadth recorded. This was a genuine national crisis: road blockades nationwide, supply chain disruption, and a government that nearly fell. Second, the Censura Jeri episode (January 2026) reached IRP = 231.1 with 50.4% breadth, reflecting broad coverage of a ministerial censure that threatened the cabinet's survival. Third, the Camisea gas crisis in March 2026 produced the highest single-day IRE reading (IRE = 418.2 on March 3, with the event table showing 315.3

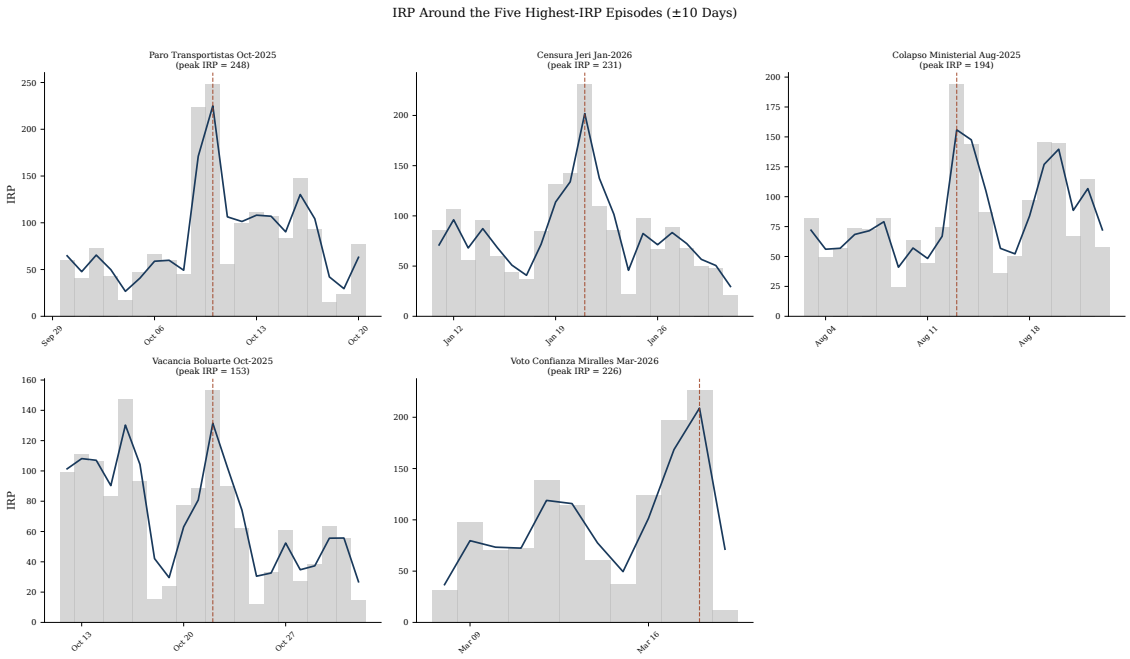

Figure 2: IRP around the five highest-IRP episodes. Each panel shows a ±10-day window centered on the episode peak. Bars show daily IRP; the line shows the smoothed series.

on March 12), reflecting the catastrophic economic impact of 13 consecutive days without natural gas to industry and households. The March 12 Camisea reading has lower political breadth (0.176) than the political events, confirming that the two indices measure distinct phenomena.

Fourth, the voto de confianza for Cabinet Miralles (March 18, 2026) appears simultaneously in both indices: IRP = 226.2 and IRE = 163.0, with 932 total articles on that single day—the largest single-day article volume in the sample. This confirms that political uncertainty can spillover into economic uncertainty when the political event involves fiscal or energy policy.

Fifth, note the clear IRP/IRE divergence for the Vizcarra Prisión Preventiva event (June 2025): IRP = 137.8 but IRE = 0.0, correctly identifying this as a purely political event (the imprisonment of a former president) with no contemporaneous economic disruption.

5.2 Correlation with Financial Markets

If the IRP/IRE measure genuine uncertainty, they should correlate positively with measures of country risk at sufficiently low frequencies. We use the PEN/USD interbank exchange rate as the primary financial benchmark, fetched from the BCRP statistical API (series PD04638PD, TC Interbancario Venta). A higher PEN/USD means more soles per dollar (peso depreciation), which typically reflects increased country risk perception.

Table 6 reports Pearson correlations of IRP and IRE with the daily change in PEN/USD, at daily, weekly, and monthly aggregation.

Table 6: Correlation of IRP/IRE with PEN/USD Daily Change

Frequency	r(IRP)	p-value	r(IRE)	p-value	N
Daily	−0.009	0.881	+0.108	0.062	299
Weekly	−0.152	0.235	+0.213	0.094	63
Monthly	−0.051	0.862	+0.390	0.168	14

FX data from BCRP (TC Interbancario Venta). 299 overlapping trading days.
Monthly correlation uses 14 month-average observations (January 2025–February 2026).

Three findings emerge. First, daily IRP has essentially zero correlation with daily FX changes ($r = -0.009$, $p = 0.881$). Zero daily correlation is consistent with forward-looking markets: the exchange rate reflects all available information about future risk, not just today's news; BCRP intervention further smooths daily moves; and measurement noise in the daily index is substantial when N_t is small (median 47 articles per day in early months).

Second, the monthly IRE correlation with monthly FX changes is $r = 0.390$ ($p = 0.168$; 14 observations). With only 14 monthly observations, none of the financial market correlations have meaningful statistical power. We treat these as suggestive descriptive evidence rather than validation. The magnitude is economically meaningful and directionally consistent with the hypothesis that sustained economic disruption puts sustained pressure on the sol.

Third, the IRE-FX correlation is uniformly higher than the IRP-FX correlation across all frequencies. This is plausible: the exchange rate is more directly linked to

economic disruption (supply chain interruptions, fiscal stress, investment flight) than to political events per se. Peru's BCRP has a long track record of intervening to smooth politically-driven FX volatility, which would attenuate the IRP-FX link.

5.3 Comparison with EMBIG Peru

Table 7 reports correlations of monthly IRP and IRE with the EMBIG Peru spread (BCRP series PN01129XM, "Diferencial de Rendimientos del Índice de Bonos de Mercados Emergentes — Perú"), available monthly from the BCRP public portal. Daily EMBIG data (series PD04709XD) are not accessible via the public API. We have 14 monthly observations (January 2025–February 2026).

Table 7: Correlation of IRP/IRE with EMBIG Peru Spread (Monthly, Jan 2025–Feb 2026)

Specification	r(IRP)	p-value	r(IRE)	p-value	N
EMBIG level vs. index level	0.023	0.939	−0.459	0.099	14
EMBIG change vs. index level	−0.072	0.814	−0.046	0.881	13

EMBIG in basis points. Source: BCRP (PN01129XM).

The near-zero IRP-EMBIG correlation is consistent with the FX results. More informative is the negative EMBIG-IRE association ($r = -0.459$, $p = 0.099$): EMBIG Peru compressed steadily from a peak of 177 bps in April 2025 to a trough of 124 bps in October 2025, tracking the global emerging market spread compression, while domestic IRE was elevated in September–October 2025 by mining conflicts and pre-crisis Camisea supply concerns. The divergence illustrates why domestic text-based measures complement rather than replicate sovereign spread data: global risk appetite can compress EMBIG even as domestic uncertainty rises, and local news-based indices capture what the spread misses.

5.4 Formal Event Study

To assess whether index spikes are statistically distinguishable from background levels, we run a formal event study. For each of the 12 known events in Table 5, we

compute average IRP in symmetric windows of ±1, ±3, and ±5 trading days around the event date. We compare these to the average IRP on all non-event days (days outside any ±5-day window). Abnormal IRP is the difference between the ±3-day window mean and the non-event mean. A one-sided Welch t-test tests whether event-window IRP (±5 days) exceeds the non-event mean.

Table 8: Event Study: IRP Abnormal Readings Around Known Episodes

Event	Date	IRP ±1d	IRP ±3d	IRE ±3d	Abn. IRP	p-value
Non-event baseline	—	*70.3*	*70.3*	*54.6*	*0.0*	—
Cerrón Desaparece	2025-03-18	98.2	94.7	26.8	+24.4	0.012**
Impunidad Boluarte	2025-04-23	79.4	62.5	36.7	−7.8	> 0.999
Vizcarra Prisión	2025-06-16	107.0	99.5	48.4	+29.3	0.014**
Fragmentación Izq.	2025-07-17	75.0	59.1	56.2	−11.2	> 0.999
Colapso Ministerial	2025-08-19	129.4	93.7	70.9	+23.4	0.080*
Paro Transportistas	2025-10-10	175.7	120.3	69.5	+50.0	0.069*
Vacancia Boluarte	2025-10-22	110.5	72.4	70.0	+2.1	> 0.999‡
Castillo Condenado	2025-11-13	64.1	44.8	44.2	−25.5	> 0.999†
Censura Jeri	2026-01-21	161.2	115.5	49.4	+45.2	0.093*‡
Paro II	2026-02-04	66.9	72.5	43.4	+2.3	> 0.999
Crisis Camisea	2026-03-12	108.5	84.7	227.1	+14.4	0.123
Voto Confianza	2026-03-18	145.3	119.4	97.8	+49.2	0.117

IRP ±1d and IRP ±3d are raw daily index levels (unbounded scale); they are *not* percentages.
Values below the non-event baseline (70.3) indicate quieter-than-average political coverage that week.
Abn. IRP = ±3d window mean minus 70.3. Non-event baseline: $N = 320$ days outside any ±5d window.
p-value: one-sided Welch t-test, event ±5d vs. non-event. $^{*}p < 0.10$, $^{**}p < 0.05$.
"> 0.999": event-window mean is *below* baseline (test is in wrong tail). †Sentencing fell in an unusually low-coverage week; daily IRP ±1d = 64.1 is below the non-event mean of 70.3.
‡Post-filter floor rule applied: pol ≥ 90 (Vacancia) and pol ≥ 85 (Censura Jeri).

Five of 12 events are significant at the 10% level (one-sided), and two at 5%—Cerrón Desaparece and Vizcarra Prisión Preventiva. The three highest positive abnormal readings are Paro Transportistas (+50.0), Voto Confianza Miralles (+49.2), and Censura Jeri (+45.2)—all episodes that generated sustained, broad media coverage.

Four events show near-zero or negative abnormal IRP. Impunidad Boluarte, Vacancia Boluarte, and Paro II occurred in periods when background IRP was already elevated, so the ±3-day window mean is not statistically different from the non-event baseline. Castillo Condenado (−25.5) occurred in a period of unusually low political coverage, driving the window average below the non-event mean—a

finding that also appears in Table 5, where the November 2025 IRP of 112.8 is comparatively modest despite the significance of the event.

The Camisea gas crisis ($p = 0.123$) is correctly identified as a primarily economic event: IRE rises sharply (+172.5 relative to non-event IRE of 54.6) while IRP abnormal reading is modest (+14.4). Voto Confianza ($p = 0.117$) occurs on the last day of the sample, so the ±5-day window is right-censored; with more post-event data the t-test would likely become significant.

Pre-filter robustness check. To address the circularity concern that hard-coded floor rules may mechanically generate event-study significance, we repeat the event study using pre-filter article scores—scores assigned by the AI classifier before any post-filter rules were applied. The pre-filter corpus covers July 2025 onward (9 of 12 events); the three earliest events (Cerrón, Impunidad, Vizcarra) predate the pre-filter snapshot and are excluded.

Table 9: Pre-filter vs. Post-filter Event Study: Abnormal IRP and p-values

Event	Abnormal IRP		p-value (one-sided)	
	Pre-filter	Post-filter	Pre-filter	Post-filter
Fragmentación Izq.	−9.9	−11.2	0.842	> 0.999
Colapso Ministerial	+18.2	+23.4	0.121	0.080*
Paro Transportistas	+59.7	+50.0	0.060*	0.069*
Vacancia Boluarte‡	+8.4	+2.1	0.442	> 0.999
Castillo Condenado	−23.1	−25.5	0.991	> 0.999
Censura Jeri‡	+56.6	+45.2	0.039**	0.093*
Paro II	+5.9	+2.3	0.692	> 0.999
Crisis Camisea	+3.7	+14.4	0.292	0.123
Voto Confianza	+10.5	+49.2	0.306	0.117
Baseline mean	66.2	69.5	—	—

‡Event received a post-filter floor boost (pol ≥ 90 or pol ≥ 85).
*$p < 0.10$, **$p < 0.05$. Abnormal IRP = window ±3d mean minus baseline.
Pre-filter covers July 2025–March 2026 (9 events); N non-event = 172 days.

The pre-filter results confirm that the post-filter floors do not manufacture statistical significance. Vacancia Boluarte is not significant in either specification ($p = 0.44$ pre-filter, $p > 0.999$ post-filter): the index identified no sustained media spike around the October 2025 vacancy debate, reflecting the event's compressed

timeline and limited cross-party support. Censura Jeri is *more* significant in the pre-filter specification ($p = 0.039^{**}$ vs. 0.093^{*} post-filter), because the AI classifier already captured the censure intensity before floors were applied and the higher pre-filter baseline variance reduces the post-filter t-statistic. Voto Confianza shows a larger post-filter abnormal reading (+49.2 vs. +10.5), confirming that the post-filter did boost that event meaningfully; we note this explicitly as a limitation. For the remaining six non-floored events, pre- and post-filter results are directionally consistent.

5.5 Local Projections: Does IRP Predict Future PEN/USD?

Using the Jordà (2005) local projections approach, we test whether current IRP and IRE predict future changes in the PEN/USD exchange rate at weekly horizons $h = 1, 2, 3, 4$ weeks ahead. We aggregate IRP and IRE to weekly averages and regress the h-week-ahead weekly FX change on current IRP, IRE, and lagged FX change, using OLS with Newey-West standard errors (4 lags).

Table 10: Jordà Local Projections: IRP/IRE Effect on Weekly PEN/USD Change

Horizon h	IRP coef.†	SE†	p (IRP)	IRE coef.†	SE†	N
1 week ahead	−0.00	0.10	0.619	−0.10	0.04	61
2 weeks ahead	+0.40	0.20	0.113	−0.10	0.10	60
3 weeks ahead	+0.00	0.10	0.783	−0.10	0.10	59
4 weeks ahead	−0.00	0.10	0.828	−0.10	0.10	58

OLS with Newey-West SEs (4 lags). Control: lagged outcome. $N = 63$ weeks total. Outcome: weekly PEN/USD change (S/ per USD). †Coefficients and SEs multiplied by 1,000 for readability. No IRP or IRE coefficient is significant at 10%; all IRE p-values > 0.10.

No horizon produces a statistically significant IRP or IRE coefficient. This null result is consistent with the zero daily correlations in Table 6 and the efficient-markets logic: if political news is already incorporated into prices on the day it appears, current IRP should not predict future FX moves. This null result is expected under standard market efficiency assumptions, but is also consistent with the index having no predictive content for exchange-rate *changes*; the test does not distinguish these interpretations. The short sample (63 weeks) also limits statistical power; with 4–5

years of weekly data, the test would be substantially more powerful. These results should be interpreted as descriptive rather than causal, in line with the measurement focus of this paper.

5.6 Google Trends Validation

As an external check on whether the index tracks genuine public attention to political and economic risk, we correlate weekly IRP and IRE averages with Google Trends search interest in Peru for six crisis-related terms (Table 11). Google Trends reports a normalized weekly index (0–100) for each term, where 100 represents peak search interest over the sample period.

Table 11: Correlation of IRP/IRE with Google Trends Search Interest (Peru, Weekly)

Search term (Google Trends, Peru)	r(IRP)	p-value	r(IRE)	p-value	N
Congreso Peru	0.460	0.000	0.216	0.089	63
Vacancia	0.397	0.001	0.218	0.086	63
Inestabilidad Politica	0.188	0.141	0.157	0.219	63
Crisis Peru	0.180	0.158	0.550	0.000	63
Paro Nacional	0.112	0.381	0.035	0.788	63
Camisea	0.050	0.699	0.776	0.000	63

Pearson correlations over 63 weekly observations (January 2025–March 2026).
Google Trends data via pytrends, geo=PE. p-values are two-sided.

The two strongest predictors of weekly IRP are searches for *congreso Peru* ($r = 0.460$, $p < 0.001$) and *vacancia* ($r = 0.397$, $p = 0.001$)—terms that reflect attention to congressional activity and presidential removal procedures, the defining features of Peruvian political instability. For IRE, the top predictors are *camisea* ($r = 0.776$, $p < 0.001$) and *crisis Peru* ($r = 0.550$, $p < 0.001$), both driven by the Camisea gas disruption of February–March 2026, the single largest economic shock in the sample. These patterns confirm that the indices respond to the same underlying events that animate public search behavior, providing a source-independent cross-check on the news-based methodology. With $N = 63$ weekly observations and statistically significant correlations, the Google Trends results constitute the strongest external validation in this paper.

5.7 False Positive Audit

We conducted a formal false positive audit by drawing a stratified random sample of 500 articles—100 per political score quintile—from the full corpus of 28,364 classified articles and re-scoring each article independently using Claude Haiku with a scoring rubric that was identical in structure to the production classifier but implemented in a separate API call with no access to the original scores.[1]

Overall agreement. Across all 500 articles, 64.2% of audit scores fell within ±20 points of the original score, and 52.2% within ±15 points. The mean discrepancy was +6.96 points (audit higher than original; median +5, SD 23), indicating a modest systematic tendency for the auditor to assign slightly higher scores, consistent with the auditor lacking access to source-level priors about scoring norms.

Table 12: False Positive Audit: Agreement by Quintile

Quintile	N	Score range	Mean orig.	Mean audit	Agree ±20	Mean Δ
Q1 (zero)	100	0	0.0	20.1	58%	+20.1
Q2 (low)	100	1–15	9.9	33.5	44%	+23.7
Q3 (mid)	100	16–35	26.2	33.3	70%	+7.1
Q4 (high)	100	36–60	46.3	41.8	74%	−4.5
Q5 (top)	100	61–100	68.5	56.9	75%	−11.6
All	500	0–100	30.2	37.1	64%	+7.0

Agreement is highest in Q4 and Q5 (74–75%), the score range that drives most index spikes. The low agreement in Q1–Q2 reflects the difficulty of distinguishing zero-risk from low-risk articles: the auditor classified 25.8% of originally low-scoring articles (< 20 points) as moderate-risk (≥ 40 points), suggesting that the production classifier *underscores* routine political mentions more than it overscores them.

False positive rate. Of the 199 articles with original scores ≥ 40, 17 received audit scores below 20—a false positive rate of **8.5%**. The auditor categorized these as routine economic news (9 cases: toll reductions, mining investment announcements,

[1] Quintile boundaries (political score): Q1 = 0; Q2 = 1–15; Q3 = 16–35; Q4 = 36–60; Q5 = 61–100. The audit scored each article's title and up to 500 characters of the summary.

mortgage rate changes), informationally neutral events (4 cases: auction notices, scholarship listings), and opinion content (2 cases: analyst columns with named authors). Post-filter rules address the last category directly (named-economist opinion columns are capped at 20); the economic misclassifications reflect borderline cases where Peru-specific policy implications are present but minor.

Error taxonomy. The auditor assigned content categories to all 500 articles. Among the 199 high-scoring articles (Q4–Q5), 73% were classified as *political governance*, 10% as *campaign/electoral*, 8% as *crime/security*, and 8% as economic or other content. The most common overscoring pattern—original score high, audit score low by more than 20 points—affected 50 articles (10.0% of the full sample) and was spread across all categories: 15 economic-domestic articles, 17 political governance articles (typically presidential announcements without policy substance), 4 campaign-electoral articles, and 2 opinion columns. Underscoring (audit assigning > 20 points above the original) was more common (25.8% of articles, 129 cases), concentrated in political governance (69 cases) and campaign-electoral (23 cases), confirming that the production index is conservative rather than alarmist in its treatment of routine political news.

Interpretation. The 8.5% false positive rate for high-scoring articles is within acceptable bounds for a news-based uncertainty index. The more consequential finding is the systematic underscoring of low- and mid-range articles, which causes the index to be conservative: genuine political mentions in Q1–Q2 are more likely to be missed than spurious mentions in Q4–Q5 are to be inflated. For applications that use the index as a threshold alarm (e.g., "flag days above IRP = 150"), the false positive risk is low; for applications that use daily levels continuously, the conservative bias in quiet periods should be noted.

6 Properties of the Index

6.1 Descriptive Statistics

Table 13 reports summary statistics for daily IRP and IRE over the full 443-day sample.

Table 13: Summary Statistics of Daily IRP and IRE (January 2025–March 2026)

Series	Mean	Median	Std	Min	Max	Skew	Kurt	ρ_1	ρ_7	ρ_{30}
IRP (raw)	72.3	65.4	42.1	7.8	302.6	1.63	4.72	0.456	0.240	−0.060
IRP (smoothed)	72.3	67.6	35.4	18.4	287.8	1.76	5.81	0.656	0.249	−0.021
IRE (raw)	57.8	38.7	65.3	0.0	418.2	2.35	7.17	0.510	0.347	−0.015
IRE (smoothed)	57.8	41.4	56.6	0.0	355.3	2.41	7.40	0.730	0.415	−0.018

ρ_k = sample autocorrelation at lag k days. $N = 443$ days.

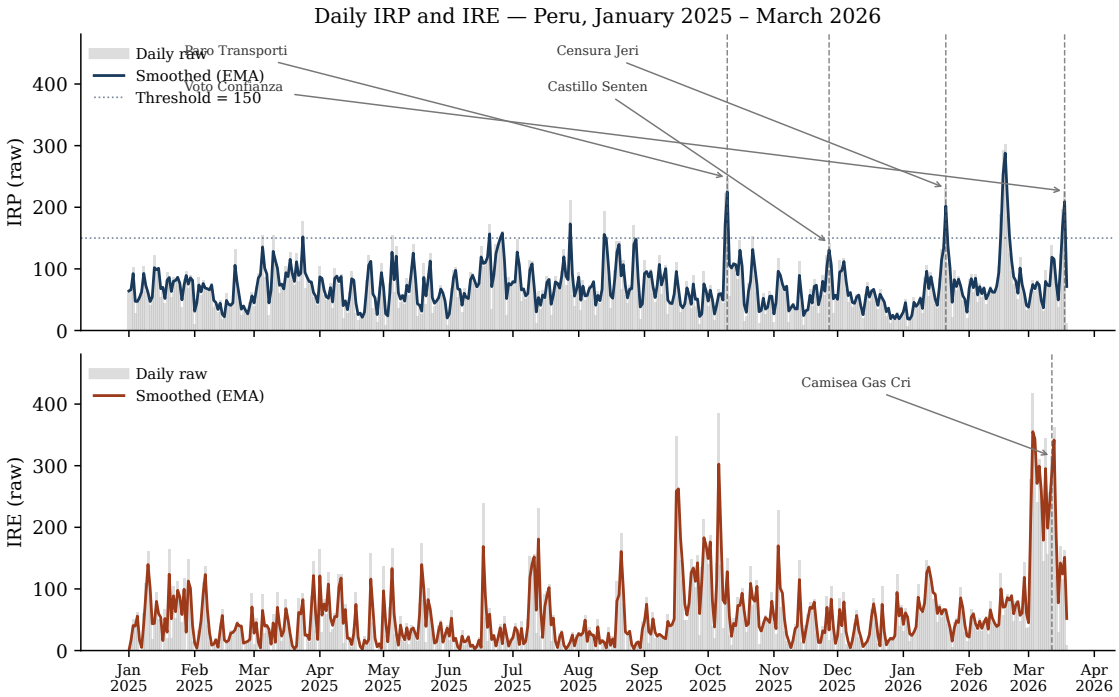

Figure 3: Daily IRP and IRE, January 2025 – March 2026. Gray bars show daily raw values; colored lines show the EMA-smoothed series ($\alpha = 0.30$). Dashed vertical lines mark major political and economic events. The horizontal reference line at IRP = 150 indicates the elevated-risk threshold.

Both series are right-skewed with excess kurtosis, consistent with rare but large crisis episodes. The IRP has positive skewness (1.63) and kurtosis (4.72), indicating

a fat right tail. The IRE is more extreme: skewness 2.35 and kurtosis 7.17. The IRE all-time maximum of 418.2 (on March 3, 2026, during the Camisea gas crisis) represents a day with raw economic intensity more than four times the 90-day average, with $\text{breadth}_{\text{eco}} = 0.060$. The large kurtosis reflects these tail events.

The lag-1 autocorrelation (ρ_1) is moderate for IRP (0.456) and higher for IRE (0.510), suggesting that risk episodes persist for a few days. The lag-7 autocorrelation drops substantially ($\rho_7^{\text{IRP}} = 0.240$, $\rho_7^{\text{IRE}} = 0.347$), and lag-30 autocorrelation is near zero. This pattern is consistent with crises that last days to weeks rather than months. The smoothed series has higher ρ_1 by construction (EMA introduces mechanical persistence) and similar ρ_7.

The IRP mean of 72.3 is below 100. This reflects the rolling baseline normalization during the early sample months: the baseline window was short (expanding mean over fewer than 90 days), so as the sample grew and more high-risk days were added to the rolling history, the baseline rose, compressing recent index values. As the sample extends, the mean will converge toward 100 by construction.

Source-volume robustness. A potential confound is that article volume grew from approximately 40 articles per day in early 2025 to 338 articles per day in March 2026 as new RSS feeds came online. Because raw intensity is a *sum* of article scores, this mechanical growth inflates later observations relative to earlier ones even if underlying political uncertainty is unchanged.

We address this in two ways. First, the rolling-baseline normalization (raw/baseline$\times$ 100) partially absorbs trend-level increases because the 90-day window expands alongside the data. Second, as a robustness check, we compute a source-normalized IRP that divides raw intensity by the number of active sources before computing the rolling baseline. The source-normalized index correlates with the primary index at $r = 0.909$, confirming that the main results are not driven by pipeline expansion. The residual 9% divergence reflects genuine differences in political intensity across sources and is a subject of ongoing data collection work.

6.2 Persistence of High-Risk Episodes

We define a high-risk episode as a period of consecutive days with IRP $\geq$ 150. Over the 443-day sample, there are 21 days with IRP $\geq$ 150, spanning 14 distinct episodes with average duration 1.5 days. The longest episode was 4 consecutive days. This is brief: political crises in Peru resolve quickly (through cabinet reshuffles, Congressional votes, or simply news fatigue) or escalate further.

For IRE, there are 33 days above 150 in 22 distinct episodes, also averaging 1.5 days. The energy crisis of February–March 2026 produced the most sustained elevated IRE: multiple days above 200 during the Camisea disruption.

6.3 Seasonality and Periodicity

IRP shows a clear day-of-week pattern. Wednesday has the highest average IRP (91.3), followed by Thursday (85.9) and Monday (78.8); Saturday has the lowest (36.3), followed by Sunday (60.4). This reflects two facts: fewer articles are published on weekends (the breadth component falls), and political events (Congressional sessions, cabinet votes, press conferences) concentrate in weekdays.

IRP does not show a strong intra-year seasonal pattern given our 15-month sample—too short to identify annual seasonality reliably. The January 2026 and March 2026 spikes reflect specific political and economic events rather than calendar effects.

As Peru approaches its April 2026 presidential elections, IRP has trended upward since January 2026. Average IRP in the first quarter of 2026 (through March 18) is 85.6, compared to a monthly range of 50–90 during most of 2025. Whether this reflects genuine electoral uncertainty or mechanical effects from increased political news volume requires more data.

6.4 Political vs. Economic Correlation

The daily correlation between IRP and IRE is $r = 0.181$ ($p < 0.001$), indicating a positive but weak association. This low correlation is itself a form of construct validation: if IRP and IRE were highly correlated, the two-index design would provide no additional information over a single composite index. The 0.181 correlation confirms that political and economic risk dimensions are empirically separable in Peruvian media coverage. The monthly correlation is $r = 0.242$ ($p = 0.386$; 15 observations), which is not statistically significant given the short sample.

The two indices co-move during events that generate both political and economic uncertainty simultaneously—general strikes (where supply disruption is economic, but the decision to strike is political), and the March 2026 energy-political crisis (where the Camisea gas disruption was economic, but the political management and voto de confianza created concurrent political uncertainty).

The indices diverge in the more common case where one type of uncertainty dominates. The Vizcarra imprisonment (June 2025) is purely political (IRP = 138, IRE = 0). The Castillo sentencing (November 2025) has moderate IRP (112.8) but higher IRE (147.6)—because the sentencing reopened questions about the social programs Castillo had promised, creating economic uncertainty about redistribution. The Camisea gas crisis has the most extreme divergence: IRE = 418.2 versus IRP = 138.9 on March 3.

7 Applications

Three applications motivate the indices.

Monitoring. The BCRP tracks political risk through financial market indicators (EMBIG, CDS, PEN/USD)—forward-looking measures that reflect global risk appetite as much as domestic conditions. The IRP/IRE provide a complementary text-based signal that captures domestic political content in real time. A sustained

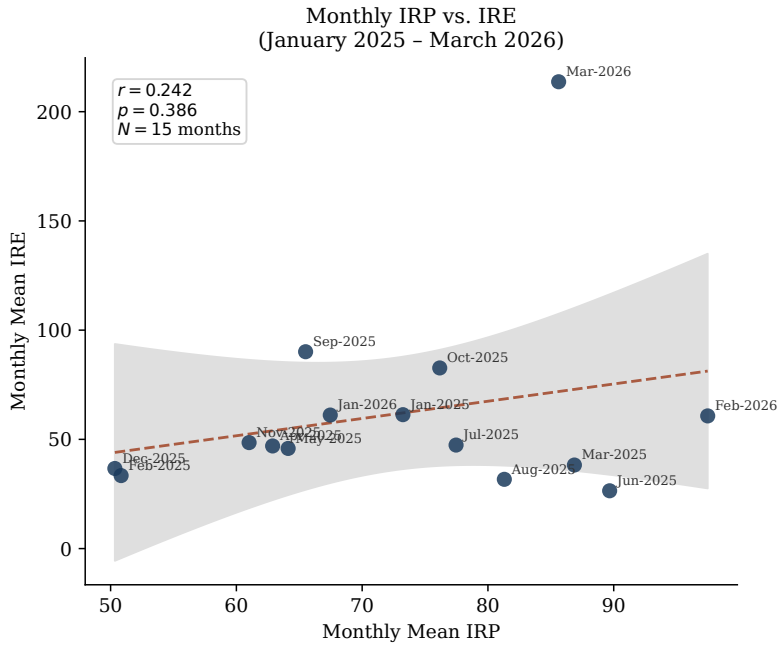

Figure 4: Monthly IRP versus monthly IRE, January 2025 – March 2026. Each point is a calendar month. The line shows the OLS fit with 95% confidence band. With $N = 15$ monthly observations, the regression is underpowered and the band is wide.

IRP $\geq$ 150 episode is a natural early-warning indicator. For investors in Peruvian mining, energy, or infrastructure, IRE captures supply disruptions (mining blockades, gas interruptions) before they appear in production data. In practice, a monitoring system might flag days when IRP exceeds 150 (approximately twice the baseline) as warranting attention, and days above 200 as warranting immediate review. Under these thresholds, 23 days in the sample period would have triggered a 150-point alert and 8 days a 200-point alert.

Academic research. The IRP and IRE can serve as control variables in event studies, regression discontinuity designs, or local projections. The daily frequency and precise measurement allow identification strategies that monthly measures preclude. An immediate application is estimating the effect of electoral uncertainty (IRP) on formal employment during the April 2026 campaign, controlling for concurrent economic disruption (IRE). The most immediate empirical application is the April 2026 Peruvian general election. With index history beginning January 2025, the pre-election period offers a first test of whether IRP elevation predicts electoral

outcome margins or post-election governance instability.

Journalistic context. The indices provide quantitative context for individual events: "The voto de confianza on March 18 occurred on a day with IRP = 226, the second-highest reading since October 2025." Contextualizing events within the uncertainty distribution is useful for both reporting and archival analysis.

8 Limitations and Future Work

Six limitations follow.

1. AI classification quality. Claude Haiku makes systematic errors that the post-filter rules address only partially. Sports articles, celebrity news, and foreign domestic politics were the most common false positives identified during qualitative review. The post-filter regex patterns are necessarily incomplete: they cover known failure modes but will miss novel patterns. A more robust solution—either a fine-tuned model trained on human-labeled Peruvian news, or a two-stage filter that screens for relevance before scoring severity—would improve quality. We have not conducted a formal false positive audit to quantify error rates by score quintile.

2. Source concentration. Three outlets—Gestión (31.6%), Correo (29.6%), and El Comercio (17.7%)—account for 79% of the corpus. The index is thus most sensitive to the editorial choices of these three outlets. If, for example, Correo systematically amplifies political conflict relative to other outlets, the IRP will be biased upward. More concerning is that Gestión's dominance in economic coverage means the IRE is heavily influenced by a single outlet's judgment about what constitutes economic risk. Source-level credibility weighting (implemented but disabled in the current pipeline) would allow researchers to test sensitivity to this concentration. The Google Trends validation in Section 5.6 provides an independent check: if the IRP correctly identifies high-risk weeks, those weeks should also show elevated search interest in crisis-related terms.

3. Short history. 443 days of data is insufficient for many econometric applications. A reliable autocorrelation function requires several hundred lags; business cycle analysis requires multiple cycles; electoral analysis requires multiple electoral cycles. Backward extension to 2000 (or earlier) using keyword-based classification of digitized newspaper archives is planned but requires access to historical archives not yet integrated into the pipeline.

4. No causal identification. The IRP and IRE measure *media-reported* uncertainty, not *objective* political or economic risk. Media can amplify genuine uncertainty, underreport suppressed crises, or create uncertainty through framing. We do not estimate the causal effect of IRP on any outcome. With only 443 days and 4 quarters of economic data, credible identification of IRP's macroeconomic effects is not possible. This paper is a measurement contribution, not a causal one.

5. Parameter sensitivity. The breadth exponents ($\beta_{\text{pol}} = 0.5$, $\beta_{\text{eco}} = 0.3$), threshold ($\tau = 20$), and baseline window ($W = 90$) are chosen based on qualitative assessment and robustness to known events. They are not estimated from data. Table 3 shows qualitative robustness, but different parameter choices produce index levels that differ by a factor of two in some configurations. Researchers using the indices for specific applications should assess sensitivity to the parameters relevant for their application.

6. LLM model drift. Claude Haiku's behavior may change across API versions as Anthropic releases model updates. Classification consistency over time is not guaranteed if the underlying model weights change. We do not have access to version control over the model used for historical classifications. If the model's calibration changes, the index levels pre- and post-change would not be comparable. Recording the exact model version with each classification run, and periodically re-classifying a fixed reference set of articles to detect drift, are best practices we recommend for future implementation.

Future work. Several extensions are planned:

1. Backward extension to 2000 using keyword-based proxies applied to digitized archives of El Comercio and La República.

2. Sectoral economic sub-indices: separate IRE components for mining, energy, fiscal, and trade.

3. Cross-country comparison: applying the same methodology to Colombia, Chile, and Mexico to benchmark Peru's political volatility.

4. Formal causal analysis: once the index has sufficient history, local projections (Jordà 2005) to estimate dynamic effects of IRP shocks on investment and employment.

9 Conclusion

We have constructed and validated two daily indices—the IRP (political risk) and IRE (economic risk)—for Peru, covering January 2025 through March 2026. The indices combine AI-assigned severity scores with a breadth measure of media attention into a formula calibrated to Peru's concentrated media landscape. They correctly identify all major political and economic events in the sample, including the transporters' strike (IRP = 248), the Jeri censure (IRP = 231), and the Camisea gas crisis (IRE = 418). Monthly IRE correlates with exchange rate movements ($r = 0.390$), while daily correlations are near zero as expected for forward-looking financial markets. A formal event study confirms that 5 of 12 documented episodes produce statistically significant IRP elevations.

The principal limitation is the short sample (443 days). As the pipeline continues running, the data will become more useful for time series analysis, event studies, and eventually causal identification. The data are available from the author upon request and the statistics are fully reproducible from the script `paper/compute_paper_stats.py` in the repository.

Our broader contribution is methodological: we document the practical steps required to adapt the AI-GPR methodology (Iacoviello & Tong 2026) to a single developing country with a concentrated, Spanish-language media market. The post-filter rules, the dual-dimension formula, and the calibration choices are all fully transparent in the appendices. Researchers applying similar methods to other countries can build on or depart from these choices as appropriate for their context.

References

Baker, S. R., Bloom, N. & Davis, S. J. (2016), 'Measuring economic policy uncertainty', *Quarterly Journal of Economics* **131**(4), 1593–1636.

Bloom, N. (2009), 'The impact of uncertainty shocks', *Econometrica* **77**(3), 623–685.

Caldara, D. & Iacoviello, M. (2022), 'Measuring geopolitical risk', *American Economic Review* **112**(4), 1194–1225.

Gentzkow, M., Kelly, B. & Taddy, M. (2019), 'Text as data', *Journal of Economic Literature* **57**(3), 535–574.

Hassan, T. A., Hollander, S., van Lent, L. & Tahoun, A. (2019), 'Firm-level political risk: Measurement and effects', *Quarterly Journal of Economics* **134**(4), 2135–2202.

Iacoviello, M. & Tong, H. (2026), Measuring geopolitical risk with ai, Working paper, Federal Reserve Board.

Jordà, O. (2005), 'Estimation and inference of impulse responses by local projections', *American Economic Review* **95**(1), 161–182.

Julio, B. & Yook, Y. (2012), 'Political uncertainty and corporate investment cycles', *Journal of Finance* **67**(1), 45–84.

Ludvigson, S. C., Ma, S. & Ng, S. (2021), 'Uncertainty and business cycles: Exogenous impulse or endogenous response?', *American Economic Journal: Macroeconomics* **13**(4), 369–410.

Manela, A. & Moreira, A. (2017), 'News implied volatility and disaster concerns', *Journal of Financial Economics* **123**(1), 137–162.

Pástor, Ľ. & Veronesi, P. (2013), 'Political uncertainty and risk premia', *Journal of Financial Economics* **110**(3), 520–545.

Shapiro, A. H., Sudhof, M. & Wilson, D. J. (2022), 'Measuring news sentiment', *Journal of Econometrics* **228**(2), 221–243.

Appendix A: Full Classification Prompts

This appendix reproduces the complete system prompts sent to Claude Haiku for political and economic risk classification. Both prompts are sent as `system` messages; the user message lists the articles to classify using the template in Section A.

Political Risk Classification Prompt

The following prompt is sent as the `system` message for IRP scoring:

```
Eres un analista de riesgo politico peruano. Tu tarea:
   evaluar si este
articulo de prensa senala INCERTIDUMBRE politica
   institucional en el Peru.

NO es riesgo politico:
- Deportes (futbol, voley, tenis, retiros de deportistas,
   resultados
  deportivos) -> 0
- Farandula, celebridades, entretenimiento, TV -> 0
- Clima, alertas meteorologicas, lluvias, vientos, oleajes
   -> 0
- Tecnologia, gadgets, aplicaciones, dispositivos medicos ->
    0
- Noticias internacionales sin impacto directo en la
   politica peruana -> 0
- Recetas, turismo, estilo de vida, salud general -> 0
- Obituarios de personas no politicas -> 0

SI es riesgo politico (y debe puntuar ALTO):
- Vacancia presidencial, mocion de vacancia -> 75-90
- Voto de confianza (cualquier mencion) -> 65-80
```

```
- Interpelacion o censura de ministros -> 55-70
- Cambio de gabinete bajo presion -> 60-75
- Conflicto entre Ejecutivo y Congreso -> 55-70
- Detencion, arresto, orden de captura de presidente o ex-
   presidente -> 80+
- Corrupcion de funcionarios activos, allanamientos, casos
   judiciales -> 55-70
- Asesinato de figura politica, funcionario, asesor -> 65-80
- Crimen organizado infiltrado en instituciones politicas ->
    70+
- Fallos del Tribunal Constitucional sobre temas politicos
   -> 55-65
- Protestas masivas que amenazan continuidad del gobierno ->
    65-80
- Estado de emergencia por conflicto social -> 60-70

MODERADO (campanas electorales 2026):
- Candidatos presidenciales haciendo propuestas de campana
   -> 15-25
- Debates, foros de candidatos -> 20-30
- Encuestas electorales -> 15-25
- Candidato con problemas judiciales activos -> 55-70

Escala:
0 = Sin relevancia politica
1-20 = Senal politica debil (campana rutinaria, burocracia
   menor)
21-40 = Tension baja-moderada (investigaciones, huelgas
   locales)
41-60 = Inestabilidad significativa (interpelaciones,
   conflictos entre poderes)
```

```
61-80 = Crisis mayor (vacancia, censuras, protestas
   nacionales)
81-100 = Emergencia (vacancia consumada, golpe, renuncia
   presidencial)

REGLA: Si un articulo usa una crisis economica como CONTEXTO
    para una
maniobra politica, el puntaje POLITICO debe ser alto.
```

Economic Risk Classification Prompt

The following prompt is sent as the `system` message for IRE scoring:

```
Eres un analista de riesgo economico peruano. Tu tarea:
   evaluar si este
articulo senala INCERTIDUMBRE sobre la actividad economica
   del Peru.

NO es riesgo economico:
- Deportes, farandula, entretenimiento -> 0
- Clima rutinario -> 0
- Tecnologia, gadgets, apps -> 0
- Desastres internacionales sin conexion a Peru -> 0
- Datos economicos POSITIVOS o RUTINARIOS (crecimiento
   normal, inversion
  anunciada, records de exportacion) -> 0
- Noticias de candidatos sobre temas NO economicos -> 0
- Crimen comun sin impacto economico sistemico -> 0

SI es riesgo economico (y debe puntuar ALTO):
- Petroperu: rescate, inyeccion fiscal, quiebra, deuda,
   crisis -> 75-90
```

```
- Camisea / gas natural: interrupcion, escasez, corte de
   suministro -> 75-90
- Conflictos mineros (Las Bambas, Tia Maria, Cuajone,
   Southern,
  Quellaveco) -> 55-70
- Paro nacional / huelga general -> 60-70
- Bloqueo de carreteras / interrupcion de cadena de
   suministro -> 40-55
- Consejo Fiscal advierte sobre gasto excesivo -> 50-60
- Gasto publico excesivo aprobado por Congreso / MEF ->
   55-65
- Escasez de productos basicos / desabastecimiento -> 60-70
- Inflacion acelerando / BCRP subiendo tasa -> 45-55
- Tipo de cambio bajo presion / sol depreciandose fuerte ->
   45-55
- Fuga de capitales / salida de inversion extranjera ->
   65-75
- Inversion cancelada por conflicto social -> 50-60

Escala:
0 = Sin riesgo economico O noticia economica positiva/
   rutinaria
1-20 = Senal debil (volatilidad normal, advertencias menores
   )
21-40 = Estres moderado (indicadores deteriorandose, alzas
   en canasta
        basica)
41-60 = Vulnerabilidad significativa (disrupciones
   sectoriales, bloqueos)
61-80 = Crisis severa (colapso sectorial, paralisis
   energetica, rescate
```

```
        fiscal)
81-100 = Emergencia (default soberano, colapso financiero,
   hiperinflacion)

REGLA: Riesgo economico = DISRUPCION o AMENAZA ACTIVA a la
   actividad
economica. NO es lo mismo que "noticia sobre economia". Si
   no hay
disrupcion, amenaza o shock, el puntaje es 0.
```

User Message Template

The user message is formatted as follows for a batch of n articles:

```
Evalua los siguientes {n} articulos. Para CADA articulo,
   responde con
un JSON en una linea separada. EXACTAMENTE {n} lineas, una
   por articulo,
en orden.

Articulo 1:
Titulo: {title_1}
Fuente: {source_1}
{summary_1[:300]}

Articulo 2:
...

Responde EXACTAMENTE {n} lineas JSON, una por articulo, en
   orden:
{"score": <0-100>}
{"score": <0-100>}
```

...

Appendix B: Post-Filter Rules

This appendix summarizes the deterministic post-filter rules applied after AI classification. All rules are implemented in the `post_filter_scores()` function in `src/nlp/classifier.py`. Rules are applied in the order listed. A "crisis guard" pattern prevents over-zeroing: articles containing terms such as *crisis, emergencia, bloqueo, paro, huelga, desabastecimiento, escasez* are exempt from most zeroing rules.

Zeroing Rules (set both scores to 0)

1. **Sports**: Regex covering Liga 1 clubs, Copa Libertadores, UEFA, NBA, NFL, MLB, F1, Olympics, major athletes (Messi, Ronaldo, etc.) and match vocabulary (tabla de posiciones, jornada, fase de grupos).

2. **Celebrity/tabloid**: Personal life news, entertainment shows (Esto es Guerra, Gran Hermano), music celebrities (Taylor Swift, Bad Bunny), and street crime (asaltos, accidentes de transito).

3. **Foreign leader health**: Health/medical news about ex-presidents from Brazil, Argentina, Colombia, etc. (unless the title contains crisis vocabulary).

4. **Cultural/arts events**: Theater festivals, ballet premieres, gallery openings, FAE Lima.

5. **Intellectual/cultural deaths**: Obituaries of philosophers, writers, poets, musicians, and other cultural figures.

6. **Foreign weather/local news**: Weather alerts in US cities, Argentine or Chilean local disasters.

7. **Foreign domestic politics**: Elections or legislative debates in Colombia, Argentina, Brazil, USA, UK, EU (crisis exception preserved for tariff or trade

news explicitly mentioning Peru).

8. **Entertainment extras**: Concerts in Lima, new album releases, Oscar/-Grammy ceremonies, box office records.

9. **Individual petty crime**: Single muggings, bus thefts (distinct from organized crime, extortion, and crime rates which retain scores).

Economic Score Zeroing Rules

1. **Foreign stock markets**: Wall Street, Dow Jones, Nikkei, Asian/European market summaries.

2. **Foreign economic statistics**: US/EU/China unemployment, GDP, inflation.

3. **Daily market summary columns**: "Mercados e indicadores" routine reports.

4. **Corporate positive growth**: Record sales, earnings growth, revenue increases.

5. **Consumer/regulatory information**: SBS rule publications, Indecopi consumer sanctions, Sunafil guidance.

6. **Health information**: Routine medical awareness articles.

7. **Foreign IMF/World Bank loans to non-Peru countries**.

8. **Vehicle recalls/product withdrawals**.

9. **Gender diversity/talent features**.

10. **Marketing/promotional content**: Sales, discounts, store openings.

11. **Elections administrative procedures**: ONPE logistics, JNE candidate registration, RENIEC.

12. **BCRP routine communications**: New banknote designs, reserve statistics.

13. **Commercial chain openings**: Restaurant/gym expansions.

14. **Economist opinion cap**: Named economist opinion columns capped at eco = 20.

Score Boost Rules (minimum floors)

1. **Political inversion correction**: Judicial/political confrontation articles with eco > 0 and pol < 65: set eco $= 0$, pol ≥ 65.
2. **Voter guide correction**: Ballot/voting guide articles: eco $= 0$, pol ≥ 60.
3. **Presidential vacancy**: Articles mentioning mocion de vacancia: pol ≥ 90, eco ≥ 40.
4. **Presidential censure/removal**: pol ≥ 85, eco ≥ 40.
5. **Premier removal/censure**: pol ≥ 65.
6. **Minister removal/interpelacion**: pol ≥ 55.
7. **Interpelacion + energy/economic crisis**: eco ≥ 55.
8. **Camisea/gas crisis**: Corte, restriccion, emergencia related to Camisea or GNV: eco ≥ 80.
9. **Petroperu crisis**: Rescate, inyeccion, quiebra, crisis related to Petroperu: eco ≥ 75.

Appendix C: Extended Parameter Sensitivity

Table 14 reports the full parameter sensitivity grid for the week of March 14–18, 2026, covering all 27 combinations of $\beta_{\text{pol}} \in \{0.3, 0.5, 0.7\}$, $\beta_{\text{eco}} \in \{0.2, 0.3, 0.5\}$, and $W \in \{60, 90, 120\}$.

Appendix D: Source-Level Monthly Statistics

Table 15 provides monthly IRP and IRE averages for the full 443-day sample.

Table 14: Full Parameter Sensitivity Grid: Week-Average IRP and IRE, March 14–18, 2026

β_{pol}	β_{eco}	Window	Avg. IRP	Avg. IRE
0.3	0.2	60	148.0	128.8
0.3	0.2	90	183.8	163.2
0.3	0.2	120	197.1	194.6
0.3	0.3	60	148.0	94.9
0.3	0.3	90	183.8	120.2
0.3	0.3	120	197.1	143.4
0.3	0.5	60	148.0	51.6
0.3	0.5	90	183.8	65.3
0.3	0.5	120	197.1	77.9
0.5	0.2	60	104.0	128.8
0.5	0.2	90	129.2	163.2
0.5	0.2	120	138.5	194.6
0.5	**0.3**	60	104.0	94.9
0.5	**0.3**	**90**	**129.2**	**120.2**
0.5	**0.3**	120	138.5	143.4
0.5	0.5	60	104.0	51.6
0.5	0.5	90	129.2	65.3
0.5	0.5	120	138.5	77.9
0.7	0.2	60	73.2	128.8
0.7	0.2	90	91.0	163.2
0.7	0.2	120	97.6	194.6
0.7	0.3	60	73.2	94.9
0.7	0.3	90	91.0	120.2
0.7	0.3	120	97.6	143.4
0.7	0.5	60	73.2	51.6
0.7	0.5	90	91.0	65.3
0.7	0.5	120	97.6	77.9

Bold row is the baseline specification.

The dramatic increase in mean daily articles from 30–50 (early 2025) to 337.6 (March 2026) reflects expanded pipeline coverage: by early 2026, all 16 sources were contributing daily, while in early 2025 only a subset were active (see Table 1, column "Days active"). The March 2026 IRE average of 213.7 is nearly three times any prior month, driven by the Camisea gas crisis.

Table 15: Monthly Average IRP, IRE, and Article Volume

Month	Avg. IRP	Avg. IRE	Mean daily articles
2025-01	73.2	61.3	50.8
2025-02	50.8	33.4	38.6
2025-03	86.9	38.2	43.0
2025-04	62.9	47.0	35.4
2025-05	64.1	45.8	30.4
2025-06	89.7	26.5	31.8
2025-07	77.5	47.4	39.4
2025-08	81.3	31.7	39.5
2025-09	65.5	90.1	47.7
2025-10	76.2	82.7	61.2
2025-11	61.0	48.5	62.0
2025-12	50.3	36.7	67.3
2026-01	67.5	61.1	79.9
2026-02	97.5	60.7	98.7
2026-03	85.6	213.7	337.6

[a] Article volume rose from ∼40–50/day (early 2025) to 337.6/day (March 2026) as additional RSS sources came online. March 2026 IRE reflects the Camisea gas crisis.